

Exercise – Understanding the theory of indirect inference

1. Aim of the exercise

The aim of this exercise is to review the theory of indirect inference as a simulation-based method for estimating parameters in models with analytically intractable likelihood functions.

2. Theory

Indirect inference is a method for estimating the parameters of a complex model when its likelihood function is difficult or impossible to work with analytically. This typically occurs in models with nonlinear relationships, unobserved variables, or missing data, where the likelihood function is either intractable or computationally prohibitive. Rather than estimating the parameters directly, the method uses simulation and comparison to approximate them.

Consider a data-generating process referred to as a complex model, also called a structural or economic model, whether derived from formal theory or specified empirically. This model is defined as

$$y_t = G(y_{t-1}, x_t, u_t; \theta),$$

where $\{y_t\}_{t=1}^T$ are observed endogenous variables, $\{x_t\}_{t=1}^T$ are observed exogenous variables, and $\{u_t\}_{t=1}^T$ are unobserved random errors. The initial condition y_0 is known, and the errors u_t are assumed to be i.i.d. according to a known distribution F . The goal is to estimate the parameter vector $\theta \in \mathbb{R}^k$ that governs this model.

To facilitate estimation, indirect inference introduces an auxiliary model, a simplified statistical representation that is easier to estimate and designed to capture salient features of the data. This model is specified as

$$y_t = h(y_{t-1}, x_t; \beta) + \varepsilon_t,$$

where the error term ε_t follows a known distribution: $\varepsilon_t \sim F_\varepsilon$. The corresponding conditional density function is given by

$$f_\varepsilon(y_t - h(y_{t-1}, x_t; \beta)) = f(y_t \mid y_{t-1}, x_t; \beta),$$

with $\beta \in \mathbb{R}^p$ denoting the auxiliary parameter vector. It is required that $p \geq k$, ensuring that the auxiliary model possesses sufficient flexibility to reflect the behavior of the complex model. Although the auxiliary model may be misspecified, its purpose is not to replicate the structural process exactly, but rather to provide a tractable summary of the key statistical features present in the data.

The auxiliary parameters $\hat{\beta}$ are estimated from the observed data by maximizing the log-likelihood function

$$\hat{\beta} = \arg \max_{\beta} \sum_{t=1}^T \log f(y_t \mid y_{t-1}, x_t; \beta),$$

yielding a statistical summary of the observed data. Simulated data are then generated from the complex model using trial values of θ . The auxiliary model is re-estimated on each simulated dataset to obtain

$$\tilde{\beta}(\theta) = \arg \max_{\beta} \sum_{t=1}^T \log f(\tilde{y}_t(\theta) \mid \tilde{y}_{t-1}(\theta), x_t; \beta),$$

where $\tilde{y}_t(\theta)$ denotes the simulated data generated under parameter θ . The core idea is to adjust θ so that the auxiliary estimates from the simulated data, $\tilde{\beta}(\theta)$, closely match those from the observed data, $\hat{\beta}$.

If the model is exactly identified ($p = k$), it may be possible to find a θ that reproduces $\hat{\beta}$ exactly. In most cases, however, the model is overidentified ($p > k$), and a distance metric is used to measure the distance between $\tilde{\beta}(\theta)$ and $\hat{\beta}$. Likelihood-based estimation of the auxiliary parameters ensures consistency and efficiency, making the comparison between observed and simulated data statistically meaningful.

In summary, indirect inference estimates the parameters of a complex model by leveraging a simpler auxiliary model. The auxiliary model serves as a lens through which the statistical behavior of the observed data is summarized and compared to simulated data. By iteratively adjusting θ to minimize the distance between auxiliary estimates from observed and simulated data, the method yields an indirect estimate of the structural parameters governing the complex model.

Indirect inference possesses well-defined asymptotic properties that support its use in econometric analysis. Under standard regularity conditions, the estimator of the parameters is consistent, meaning it converges in probability to the true parameter value as the sample size increases. Furthermore, the estimator is asymptotically normal, which allows researchers to construct confidence intervals and perform hypothesis tests using standard statistical tools. These properties hold even though the auxiliary model may be misspecified, as long as it captures relevant features of the data and the mapping from structural to auxiliary parameters is sufficiently informative. The asymptotic behavior is driven by the sample size, not by the number of simulations, which primarily affects the precision of the estimator in finite samples.

3. Final notes

This file is prepared and copyrighted by Simonas Stravinskas and Tunga Kantarcı. This file and the accompanying MATLAB file are available on GitHub and can be accessed using this [link](#). This file has made use of Osborn, 1976. Maximum likelihood estimation of moving average processes. *Annals of Economic and Social Measurement*, 5 (1), 75-87.